

M102 型售货机控制卡

M102 Vending Machine control board

使用说明 **instructions**

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2. 版本修订

2.Revision

2018-9-30 全新制定

2018-9-30 new revision

3. 产品概览

3.Product overview

-  意法半导体芯片控制
- S.D. semiconductor chip control
-  精密逻辑控制电路
- Precision logic control circuit
- 矩阵式电路结构,板载矩阵交叉电路, 减少线束成本
- Matrix circuit structure, on-board matrix cross circuit, reduce the cost of wiring harness
- 支持 100 通道的出货机构驱动, 60 电机 40 电磁锁
- Supports 100-channel shipping mechanism drive, 60 motor 40 electromagnetic lock
- 三线电机和两线电机混合使用
- The three-wire motor is mixed with the two-wire motor
- 电机与电磁铁混合使用
- The motor is mixed with the electromagnet
- 支持最多 255 块控制卡并联扩展, 实现无限扩展
- parallel extension support up to 255 pieces of control card, realize the unlimited extension
- 增强型灵活通讯口, 板载 TTL,RS-232、RS-485 接口
- Enhanced flexible communication port, on-board TTL, RS-232, RS-485 interface
- 高抗干扰设计
- High anti-interference design
- 掉货检测接口
- Drop detection interface
- **开关量输出: 8 路开关量输出**

- Switch output: 8 switch output
- 开关量输入：8 路开关量输入
- Switch quantity input: 8 switch quantity input
- 温度采样： 1 路温度采样
- Temperature sampling: 1 way temperature sampling

4. 电气特性说明

4. Description of electrical characteristics

供电电压: **DC24V** (12V 可选)

Supply voltage: DC24V (12V optional)

静态电流: 小于 **20mA**

Static current: less than 20mA

驱动电流: 最大 **1A**

drive current: Maximum 1A

通讯接口: **RS-232**

communication interface: RS-232

1) 供电电源

1) Power supply

供电电压: DC24V (可定制 DC12V 版本)。电源功率不低于 50W。

Supply Voltage: DC24V (customizable DC12V version). The power supply is not less than 50W

2) 负载

2) Load

支持 3 线制马达 80 个

Support 3-wire motor 80

支持 2 限制马达 100 个

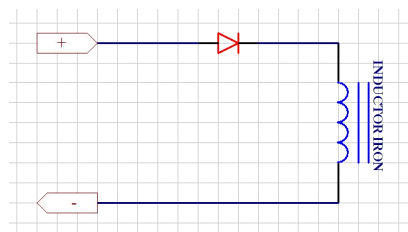
Support 2 limit motor 100

支持 电磁锁 100 个

Support electromagnetic lock 100

a) 无反馈电磁铁, 可以驱动 100 路

a) No feedback electromagnet, which can drive 100 channels

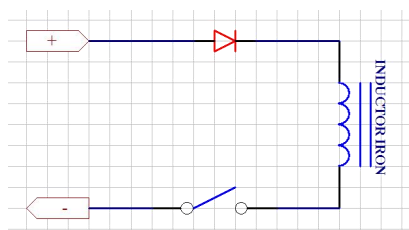


b) 有反馈电磁铁, 可以驱动 100 路

当电磁锁关闭时, 开关处于闭合状态; 当门弹开后, 开关处于断开状态

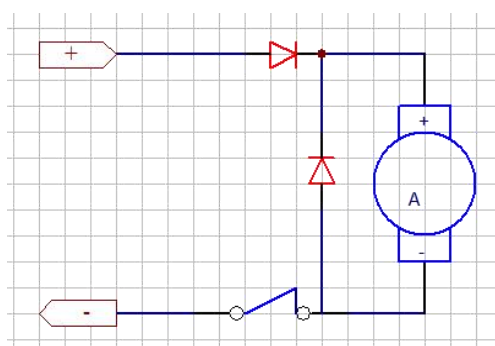
b) It has feedback electromagnet, which can drive 100 channels

When the electromagnetic lock is closed, the switch is in a closed state; When the door pops open, the switch is on



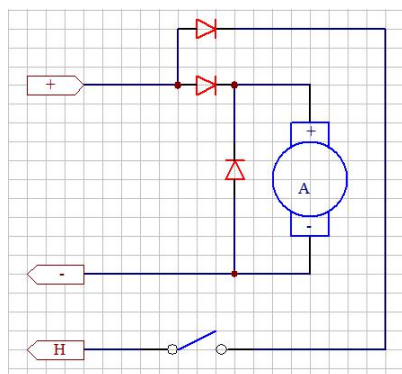
c) 两线制电机，可以驱动 100 路

c) two-wire motor that can drive 100 channels



d) 三线制电机，可以驱动 80 路

d) Three-wire motor, can drive 80 channels



3) 开关量输出

3) Switch output

带有 8 路开关量输出回路，系统 id 为 0~7。对外输出 0V 或者 高阻抗。支持吸入电流 50mA 每路（足够驱动普通 24V 继电器）。

With 8 switch output circuit, the system id is 0 to 7. Output 0V or high impedance. Supports suction current of 50mA per pass (enough to drive

normal 24V relays)

4) 开关量输入

4)Switch input

有 8 路开关量输入回路，编号为 0~7。外部可以输入 0~24V 或者开关按钮信号。

There are 8 switch input circuits, numbered 0~7.External input can be 0~24V or switch button signal.

5) 测温回路

5)Temperature measurement circuit

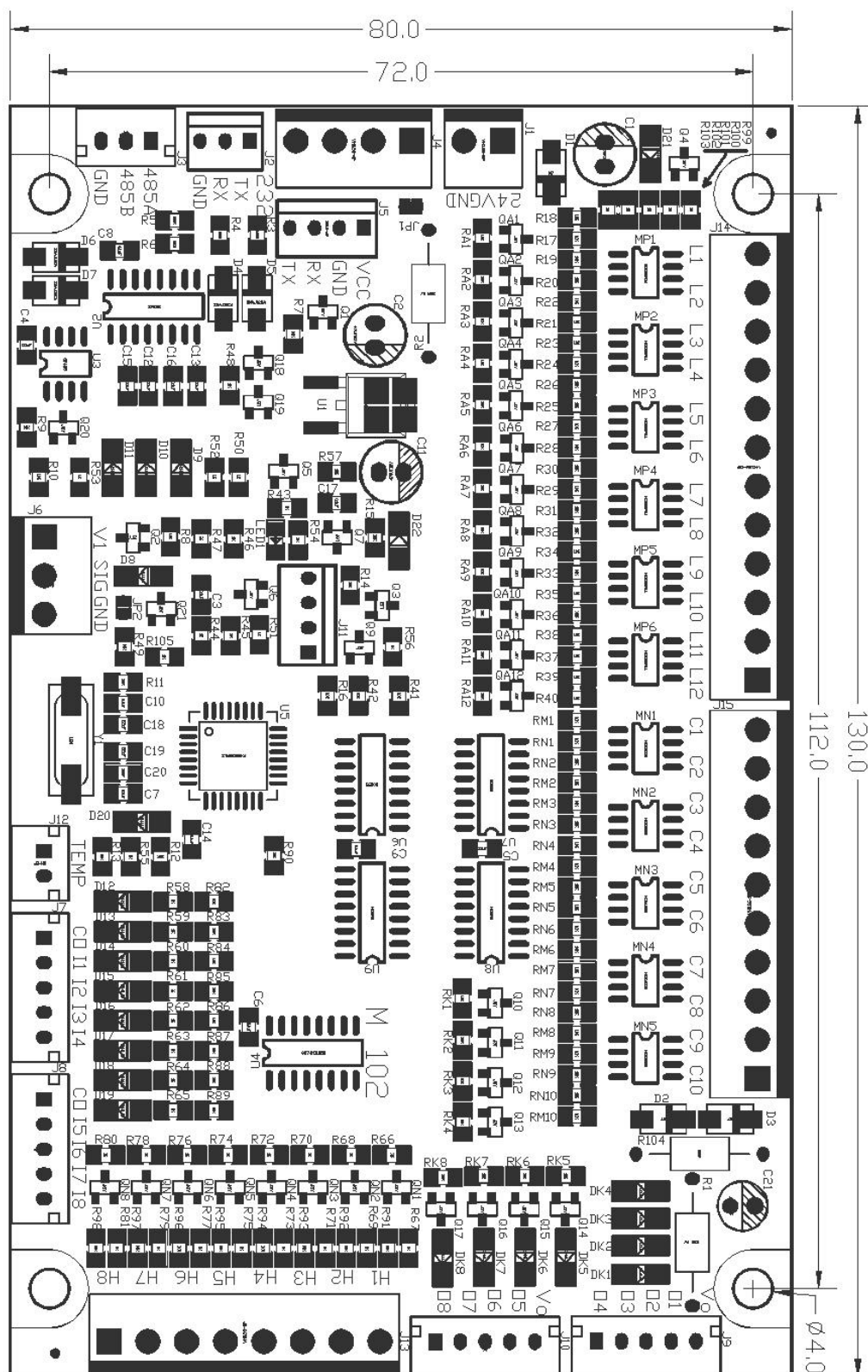
带有 1 路测温回路，配套传感器型号为 NTC 10K@25℃ B=3950。温度测量范围为-50.0℃~50.0℃，未接传感器时系统返回-50.0℃。

With 1 temperature measuring loop, the supporting sensor model is NTC 10K@25℃ B=3950.The temperature measurement range is -50.0℃~50.0℃, and the system returns -50.0℃ without the sensor.

5. 安装尺寸 5. Installation dimensions

单位(mm)

Unit (mm)

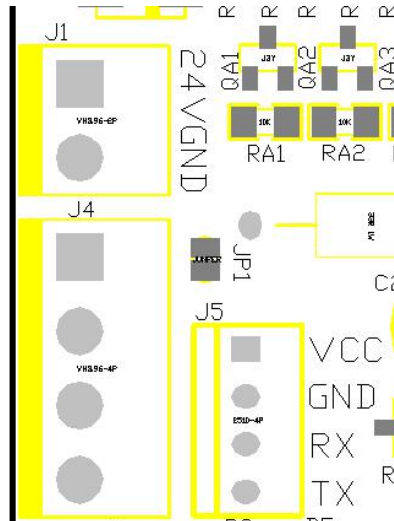


6.接线方式

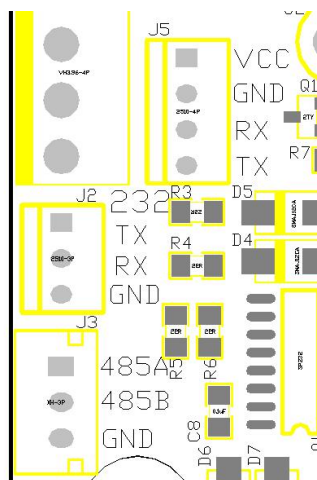
6.Connection mode

6.1 电源和通讯线的连接

6.1 Connection of power supply and communication cable



- (1) 供电电压 24V，12V 可选
- (2) Supply voltage 24V,12V optional
- (3) 静态电流 小于 40mA
- (4) Static current less than 40mA
- (5) 最大电流取决于负载电流
- (6) The maximum current depends on the load current
- (7) J1 供电
- (8) J1 power supply
- (9) J4 为了兼容 M101G
- (10) J4 for compatibility with M101G
- (11) J5 为了兼容 M101H\M101J
- (12) J5 for compatibility with M101H\M101J



J5 TTL 通讯接口

J5 TTL communication interface

J2 RS-232 接口

J2 RS-232 interface

J3 RS-485 接口

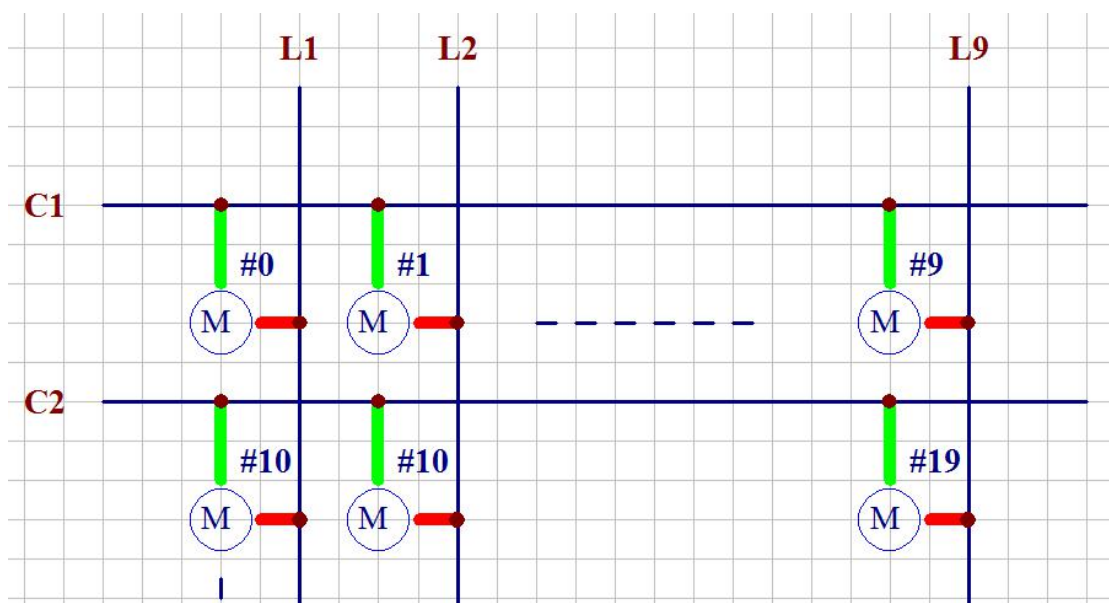
J3 RS-485 interface

6.2 电磁铁或两线制电机接线方式

6.2 Electromagnet or two-wire motor connection mode

两线制接口需要使用到的信号是 L1~L10 ; C1~C6 , C7~C10 。 L 信号输出 24V 正极, C 信号输出 0V。L 与 C 之间接负载形成回路。

The signals needed for the two-wire interface are L1~L10;C1 ~ C6 and C7 ~ C10.L signal output 24V positive pole, C signal output 0V.The indirect load between L and C forms a loop.



图中的电机，实际是安装过选通二极管、续流二极管、位置检测微动开关的组合，并非普通电机。

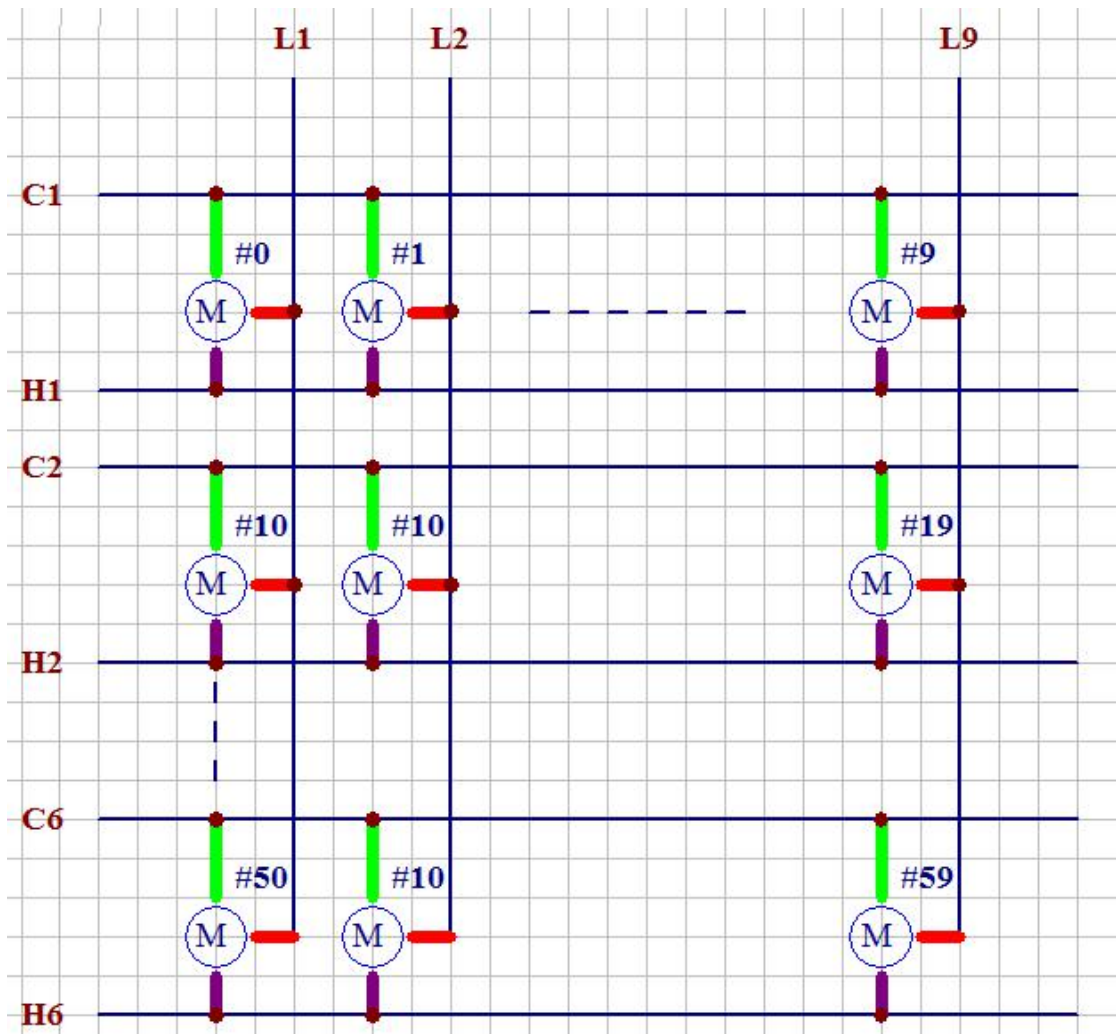
The motor shown in the figure is actually a combination of the over-gating diode, the continuous-flow diode and the position detection micro switch, rather than an ordinary motor.

6.3 三线制电机接线方法

6.3 Three-wire motor wiring method

三线制接口需要使用到的信号是 L1~L10 ; C1~C6 ; H1-H6 。L 信号输出 24V 正极，C 信号输出 0V。L 与 C 之间接负载形成回路。H 信号是电机到位的检测信号，当电机停止时是高阻抗，当电机运转过程中是 24V。

The signals required by the three-wire interface are L1~L10;C1 ~ C6.H1 - H6.L signal output 24V positive pole, C signal output 0V.The indirect load between L and C forms a loop.H signal is the detection signal when the motor is in place, high impedance when the motor stops, and 24V when the motor is running.



图中的电机，实际是安装过选通二极管、续流二极管、位置检测微动开关的组合，并非普通电机。

The motor shown in the figure is actually a combination of the over-gating diode, the continuous-flow diode and the position detection micro switch, rather than an ordinary motor

6.4 光幕接线方式

6.4Light curtain wiring Mode

光幕接口 信号 V1; SIG;GND。 V1 给光幕供电, SIG 光幕信号, GND 是光幕的电源地线。

Light curtain interface signal V1;SIG; GND。 V1 supplies power to the light curtain, SIG light curtain signal, GND is the power ground of the light curtain.

待机时: V1 无输出, 光幕不供电。 此时 SIG 需要是, 高阻抗或者断路。

Standby: V1 no output, light curtain no power supply.SIG needs to be high impedance or open circuit.

启动马达前, V1 输出 24V 给光幕供电, 此时 SIG 需要是 0V (或者说与 GND 通路)。货物掉落经过光幕时, SIG 信号是有高阻抗或者断路。

Before starting the motor, V1 outputs 24V to supply power to the light curtain, at which point SIG needs to be 0V (or GND channel).The SIG signal is high impedance or open when the cargo falls through the light curtain.

6.5 温度检测接线

6.5Temperature detection wiring

TEMP 接线, 仅需要接上, 规格符合热敏电阻即可。

TEMP wiring, only need to be connected, the specification conforms to thermistor.

6.6 开关量输出端口

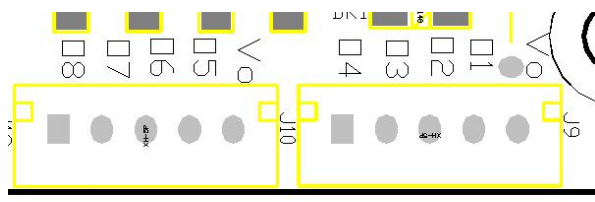
6.6Switch output port

Vo 输出 24V

Vo 24 v output

O1, O2, O3, O4 ,O5,O6,O7,O8 吸入电流 (负载能力 40mA) 可以驱动 24V 继电器, 驱动光耦时需要串联 10K 1/4W 电阻。

O1, O2, O3, O4,O5,O6,O7,O8 suction current (load capacity: 40mA) can drive 24V relay. To drive optocoupler, 10K 1/4W resistor is required in series

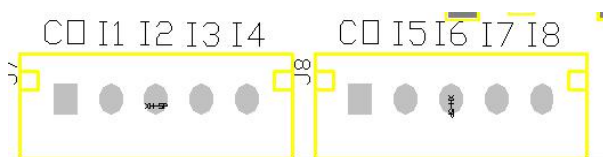


6.7 开关量输入端口

6.7Switch input port

CO 端口是公共线，I1，I2，I3，I4,I5,I6,I7,I8 是每路的输入点，可以接入外部开关信号

CO port is the common line, I1, I2, I3, I4,I5,I6,I7 and I8 are the input points of each channel, which can access external switching signals



7.通讯参数

7.Communication parameters

控制卡采用 TTL 串口方式进行通讯,可以方便转换成 232 或者 485 电平。

The control card communicates by means of TTL serial port, which can be easily converted to 232 or 485 levels.

通讯参数: 9600bps, 1 停止位, 无奇偶校验, 无流量控制

Communication parameters: 9600bps, 1 stop bit, no parity, no flow control

在通讯过程中, 上位机 (安卓或者 PC, 称为主机) 与控制卡 (称为从机) 采用主从方式

In the communication process, the host computer (Android or PC, known as the host) and the control card (known as the slave) adopt the master-slave mode

1) 每次通讯由主机发起请求, 从机应答

1) Each communication shall be requested by the host and answered by the slave

2) 主机收到上次请求的应答 或者超时后才可以发起新的请求

2) The host can only issue a new request after receiving the reply or timeout of the last request

3) 在一个通讯网络中可包含一台主机和若干台从机

3) A communication network may contain one host and several slaves computers

4) 在一个通讯网络中可包含一台主机和若干台从机, 通过个设备地址(1 字节), 来区分这些设备。其中主机的地址为 0, MS101G 控制卡的地址为 1-8。

4) A communication network may contain one host and several slaves, which can be distinguished by the address of each device (1 byte).The host address is 0, and the MS101G control card address is 1-8.

5)此外地址 255 作为广播地址。目前此地址只用作从机地址修改。**(慎重使用)**

5) In addition, address 255 serves as the broadcast address.This address is currently used only for slave address modification.**(Use with caution)**

6)控制卡卡上的指示灯的连续闪烁次数即为它的通讯地址。

6)To control the continuous flashing times of the indicator light on the card

is its address.

通讯流程如下:

The communication process is as follows:

- 1.主机下发指令
- 1.The master gives instructions
- 2.等待 50ms
- 2.Wait for 50 ms
1. 接收从机响应(超时 1 秒)
- 3.Receive slave response (timeout 1 second)

8.指令格式

8.Instruction format

为了便于上位机程序的编写，通讯协议中使用了定长数据帧，一帧指令或响应的长度为 20 字节，包含了地址，指令,数据段及校验代码四部分。当数据段中有效数据小于 16 字节时，需要填 0 补充至 16 字节。

In order to facilitate the programming of upper computer, fixed-length data frame is used in the communication protocol. The length of a frame instruction or response is 20 bytes, including address, instruction, data segment and verification code. When the valid data in the data segment is less than 16 bytes, you need to fill in 0 to supplement 16 bytes

格式如下:

Format as follows:

指令: 主机 -> 从机

Instruction: host -> slave

从机地址 (1byte)	指令 (1byte)	数据 (16byte)	校验代码(2byte)
Slave Address(1byte)	Instruction(1byte)	Data(16byte)	Check Code(2byte)

数据用 Y1~Y16 表达

响应: 从机->主机

Response: slave -> host

主机地址 (1byte)	指令 (1byte)	数据 (0-16byte)	校验代码(2byte)
Master Address(1byte)	Instruction(1byte)	Data (0-16byte)	Check Code(2byte)

数据用 Z1~Z16 表达

The data is expressed in Z1 to Z16

地址:

Address

从机地址 1~8，主机地址为 0。

指令:

Instructions:

详见指令表

See instruction table for details.

数据:

Data

数据段可包含 16 字节，指令中的数据用 Y1-Y16 表达，响应中的数据用 Z1-Z16 表达。数据段中可包含单字节和双字节两种整数，其中 16 位整数的高字节在前，低字节在后。

The data segment can contain 16 bytes, and the data in the instruction is expressed as Y1-y16, and the data in the response as Z1-Z16.A data segment may contain both single-byte and double-byte integers, with the high byte of a 16-bit integer coming first and the low byte following.

校验代码:

Check code

采用 CRC16 校验方式，运算过程见附录。校验代码的低字节在前、高字节在后。

CRC16 calibration method is adopted, and the operation process is shown in appendix.The low byte of the validation code comes before the high byte.

9.详细指令模式

9.Detailed instruction mode

5.1 获取序列号(ID) 指令:01H

5.1 Obtain sequence number (ID) instruction :01H

Z1-Z12	控制卡的序列号	
Z1-Z12	The serial number of the control card	

5.2 查询 Run 执行状态(Motor Poll)指令:03H

Z1	Run 指令执行状态	0 = 空闲 1 = 执行中 2 = 执行完毕
----	------------	-------------------------------

Z1	Run instruction execution	status 0 - Idle 1 - Execute 2 - Executed
Z2	正在运转的电机编号	00~59（共 60 个）
Z2	Running motor numbers	00 to 59 (60 in total)
Z3	执行结果	0 = 成功。 1 = 过流。负载过重,或者卡货 2 = 欠流。一般是电线断开,或者负载未安装 3 = 超时。指超过 7 秒仍未检测到电机到位信号。（一般是由于负载过重、卡货或开关电源干扰引起） 4 = 光幕自检失败，未启动电机。 5 = 有反馈电磁铁门未开
Z3	The execution result	0-Successful 1-Overflow Heavy load, or cargo stuck 2-underflowTypically,the wire is disconnected, or the load is not installed 3-timed out Means that the motor in place signal has not been detected for more than 7 seconds. (Generally due to heavy load, stuck goods or switching power interference) 4-Light curtain self-test failed and the motor was not started 5 - The feedback electromagnetic door is not open
Z4-Z5	峰值电流	单位 mA
Z4-Z5	peak current	Unit mA
Z6-Z7	平均电流	单位 mA
Z6-Z7	Average	Unit mA
Z8-Z9	运行时间	单位 ms
Z8-Z9	Running time	Unit ms
Z10	光幕状态	0=无掉货 1~200 表示货物经过光栅的时间（单位 ms）
Z10	Light curtain state	0 - No drop 1 to 200 indicates the time the goods pass through the grating (unit ms)

9.3 测试(Motor Scan)指令:04H

9.3 Test(Motor Scan)Instruction:04H

Y1	电机索引号	00~99
Y1	Motor index number	00~99
Z1	结果	AA=正常 BB=异常 CC=过载

Z1	Result	AA= normal BB= abnormal CC= overload
----	--------	--------------------------------------

9.4 启动电机(Motor Run)指令: 05H

9.4 Motor Run instruction: 05H

Y1	电机索引号	00~59
Y1	Motor index number	00~59
Y2	电机类型	0=无反馈电磁铁 1=有反馈电磁铁 2=两线制电机 3=三线制电机
Y2	The type of motor	0= No feedback electromagnet 1= Feedback electromagnet 2= Two-wire motor 3= three-wire mot
Y3	光幕工作模式	0 = 电机运转不参考光幕。 1= 启动电机前，需光幕自检成功；电机到位后停止。 2 =启动电机前，需光幕自检成功；当光幕检测到掉货后电机立即停止，未检测到掉货时电机，一直运转直到超时（7s）
Y3	Light curtain operating mode	0 = The motor operation does not refer to the light curtain. 1= Before starting the motor, the light screen should be self-checked successfully;Stop the motor when in place. 2 = Before starting the motor, the light screen should be self-checked successfully;When the light curtain detects the goods falling off, the motor will stop immediately. When the goods falling off is not detected, the motor will continue to operate until timeout (7s).
Y4	电机过流阈值	1-255 代表 10mA-2550mA。 0 表示默认值()
Y4	Motor overflow threshold	1-255 represents 10MA-2550 mA. 0 is the default value ()
Y5	电机欠流阈值	1-255 代表 10mA-2550mA。 0 表示默认值()
Y5	Motor undercurrent	1-255 represents 10MA-2550 mA. 0 is the default value () threshold

Y6	电机到位超时时间	1-100 代表 0.1S – 10.0S。 0 表示默认值(7 秒)
Y6	Motor in place time out •	1-100 represents 0.1s -- 10.0s. 0 represents the default value (7 seconds)
Z1	指定执行结果	0=已启动; 1=无效的电机索引号; 2=另一台电机在运行;
Z1	Specify the execution result	0= Started; 1= Invalid motor index number; 2= Another motor is running;

9.5 读温度(Read Temp)指令 07H

9.5Read Temp instruction 07H

本驱动板卡带有一路温度采样回路

This drive board card has a temperature sampling loop all the way

Z1-Z2	温度值	-50.0℃~50.0℃，未安装传感器时为-50.0℃。
Z1-Z2	temperature value	-50.0 to 50.0 degrees C, -50.0 degrees C when the sensor is not installed

FE0C = -500 代表-50.0

FE0C - -500 for -50.0

01F4 = 500 代表 50.0 读

01F4 s 500 for 50.0 reading

9.6 开关量输出(Write DO)指令 08H

9.6 Switch output (Write DO) instruction 08H

本驱动卡带有 4 路开关量输出回路，编号为 0~3。

This drive card has a 4-way switch output circuit, numbered 0 to 3.

Y1	DO 索引	0-6
Y1	The DO index	0-6
Y2	操作	1=ON 0=OFF
Y2	Operation.	1=ON 0=OFF
Z1	DO 索引	0-6
Z1	The DO index	0-6
Z2	操作结果	Z2=Y2+F0 . 00->F0 01->F1
Z2	The result of the operation	Z2=Y2+F0 . 00->F0 01->F1

9.7 读开关量状态(Read DI)指令 09

9.7Read switch state (Read DI) instruction 09

本驱动卡带有 4 路开关量输入回路，编号为 0~3。

The drive cassette has 4 switch input circuits, numbered 0~3.

主机下发			
The host issued			
从机地址	指令	数值	校验代码
The address from the master	Instructions	Numerical.	Check code
01~08	09	Y1~Y16	CRC-16
01~08	09	Y1~Y16	CRC-16

Y1=0

从机应答			
主机地址	指令	数据	校验代码
Answer from the slave	Instructions	Numerical.	Check code
01~08	08	Z1~Z16	CRC-16
01~08	08	Z1~Z16	CRC-16

Z1~Z4 ==DI1~DI4 1=外部连接 0=外部断开

Z1~Z4 ==DI1~DI4 1= external connection 0= External disconnect

9.8 设置板卡地址 Set Address 指令 FF

9.8 Set the board Address Set Address instruction FF

该指令用于修改驱动卡的通讯地址。按以下步骤操作：

This instruction is used to modify the address of the driver card.Follow these steps:

整个网络中只保留主机和一块驱动卡。

1)Only the host and one driver card are kept in the whole network.

发送设置设置指令，人工确认指示灯 闪烁次数。

2)Send setting instructions and manually confirm the blinking times of the indicator light.

该网络恢复原样。

3)The network is restored to its original state

注意：该指令为的通讯地址采用广播地址（255）。

Note: this instruction uses a broadcast address for the correspondence address(255)

板卡地址 可以根据指示灯闪烁次数获得

The board address can be obtained based on the number of times the light flashes

Y1	驱动卡新地址	1-8
Y1	Driver card new address	1-8
Z1	驱动卡新地址	1-8
Z1	Driver card new address	1-8

10.样例

10.Sample

获取ID

Get the ID

01 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 71 88
00 01 00 64 00 3B 04 47 36 32 33 38 36 39 00 00 00 00 6F 8E

启动#0 号马达 3 线制 无光幕

Start #0 motor 3 wire matt curtain

01 05 00 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 70 48
00 05 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 62 B5

获取马达旋转结果

Get the result of motor rotation

01 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 D0 E8
00 03 02 00 00 05 BA 00 77 0A 3C 00 00 00 00 00 00 00 00 98 46

Z1=02 运转完毕

Z1=02 run out

Z2= 00 当前操作 id 0

2= 00 The current operation id is 0

Z3=00 无故障

Z3 = 00 trouble-free

Z4,Z5=05BA 峰值电流 1466mA

Z3=00 fault-free Z4,Z5=05BA peak current 1466mA

Z6,Z7= 0077 平均电流 119mA

Z6,Z7= 0077 average current 119mA

Z8,Z9=0A3C 运转耗时 2620mS

Z8,Z9=0A3C operation time 2620mS

修改从机地址

Modify slave address

```
FF 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 20 50
FF 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 D0 A0
FF 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 81 30
FF 04 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 33 01
FF 05 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 62 91
FF 06 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 92 61
FF 07 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 C3 F1
FF 08 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 F6 02
```

启动 #1 路 DO

Start #1 DO

```
01 08 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 66 DA
00 08 00 F1 00 00 00 00 00 00 00 00 00 00 00 00 00 00 C7 62
```

关闭 #1 路 DO

Close #1 DO

```
01 08 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 A7 DA
00 08 00 F0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 06 62
```

读取 DI

Read DI

```
01 09 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 F6 4A
00 09 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 A7 B6
```

11.使用流程

11.Using process